



Faculty of Computers and Artificial Intelligence Cairo University

IT351: Information Theory and Data Compression

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Assignment 1 (LZ77)

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This document analyzes the **Best Case, Average Case, and Worst-Case** time complexity of the implemented LZ77 compression and decompression algorithms.

1. Compression

The compression algorithm searches for the longest matching sequence inside the sliding window. For every current position, it checks possible starting positions in the window and compares matching characters.

Case	Time Complexity	Explanation
Best Case	$O(n)$	There are little or no useful matches, so the algorithm finishes each position with only a small amount of work.
Average Case	$O(n^2)$	The algorithm may examine several positions in the window and perform character comparisons for different match lengths.
Worst Case	$O(n^2)$	Highly repetitive input such as AAAAAAAAAAAAAA... can cause long character comparisons for multiple candidate positions.

Note: WINDOW_SIZE is 6 in this implementation. If the window size is treated as a fixed constant and the matching operation is bounded accordingly, the search over the window can be considered constant per position. However, in the current implementation the inner matching loop can grow with the input length, so $O(n^2)$ is the safe worst-case analysis.

2. Decompression

The decompression algorithm processes each compressed token and copies the required number of characters into the output message. The

copying is done character by character, which also supports overlapping matches.

Case	Time Complexity	Explanation
Best Case	$O(n)$	The output message contains no characters, and each character is processed.
Average Case	$O(n)$	The algorithm processes the characters that form the decompressed message.
Worst Case	$O(n)$	Even with long overlapping matches, each output character is generated once.

3.Summary

Algorithm	Best	Average	Worst
Compression	$O(n)$	$O(n^2)$	$O(n^2)$
Decompression	$O(n)$	$O(n)$	$O(n)$

Here, n represents the size of the input/output message. The complexity describes the implemented code, including its nested matching loop.